

ECEN 665

Section – 601

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TA: Dharma Paladugu
RF Frontend Integration

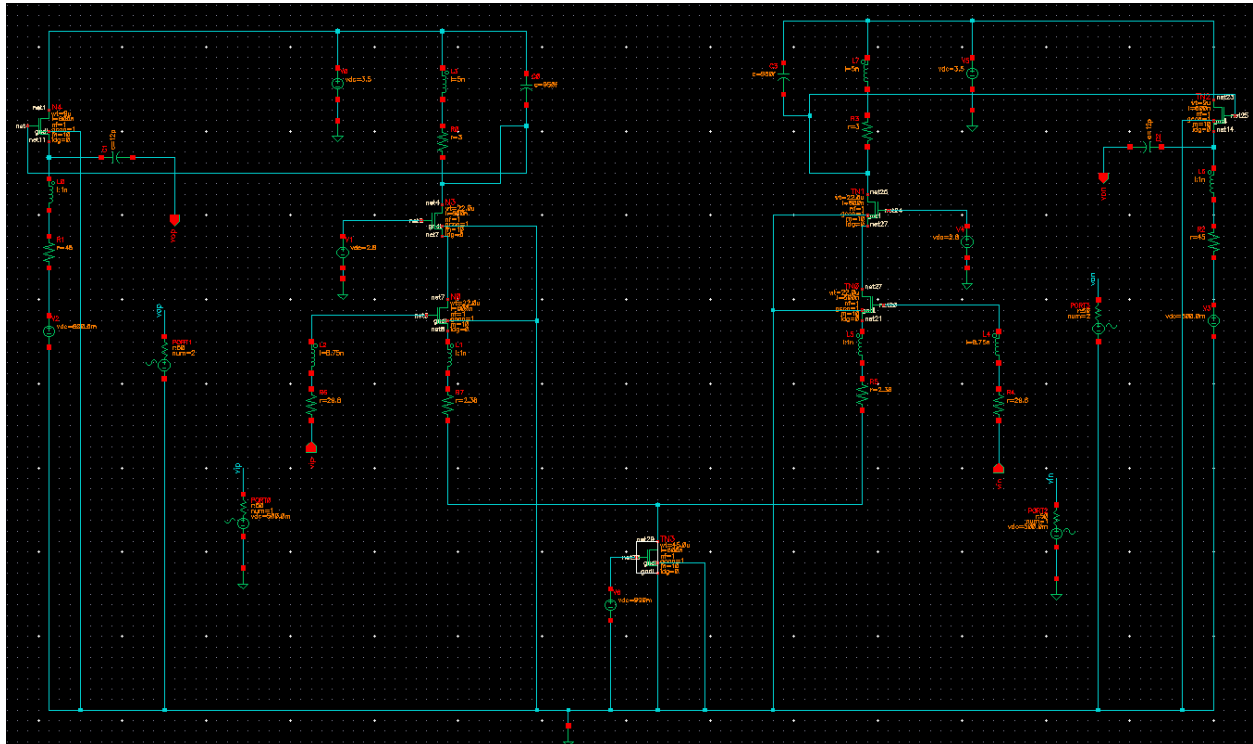
Lab 06: (Post-lab) report

Date: 26-04-2026

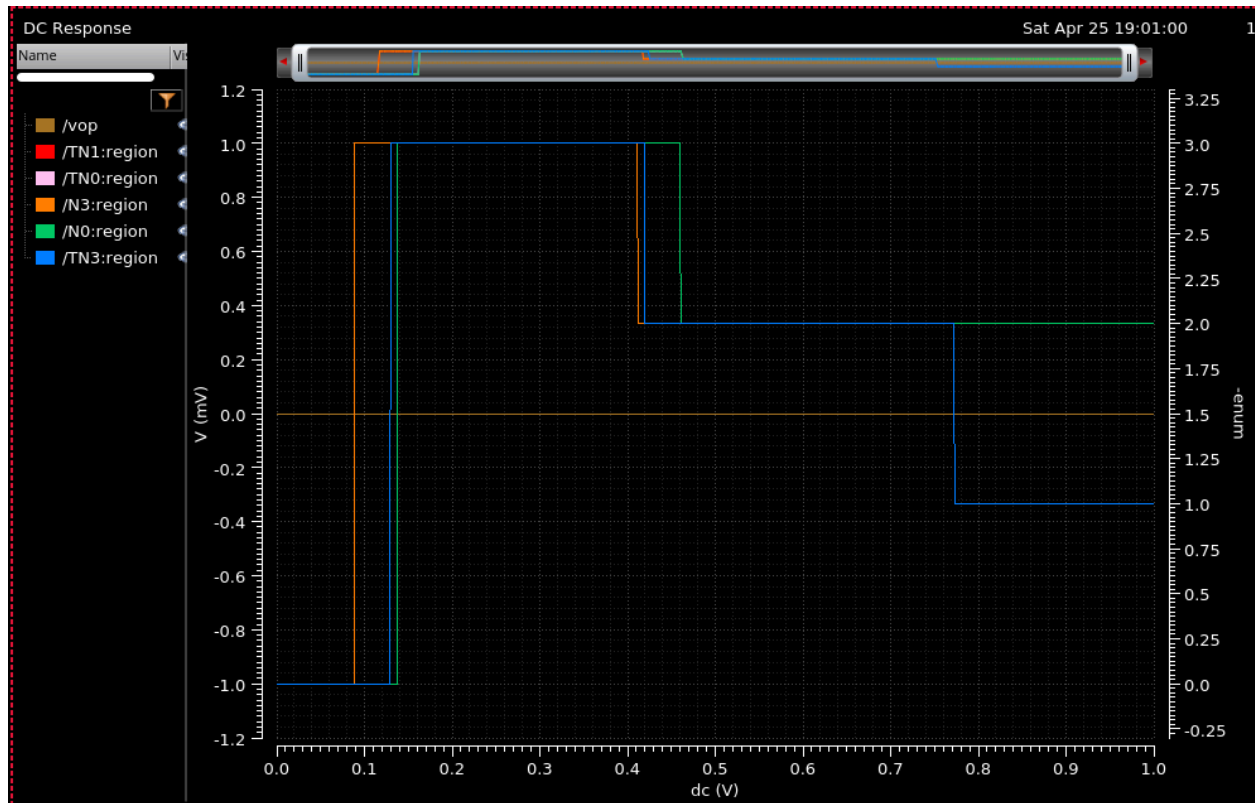
Texas A&M University

1. LNA re-design: Starting from your single-ended LNA from Lab 4, remove the output buffer and create a fully differential version. The ground of the single-ended LNA now becomes a virtual ground at the drain of a NMOS transistor which serves a current source. The tail current should be twice the current of the single ended LNA. Notice that you will have to adjust the bias voltage of the input and cascode transistors to make sure that the transistor at the current source is in saturation. The LNA will now be matched to 100 ohms differential. You can choose to continue with this new impedance or adjust the matching network to make the differential input of the LNA match 50ohm (extra credit).

1. Making it differential:



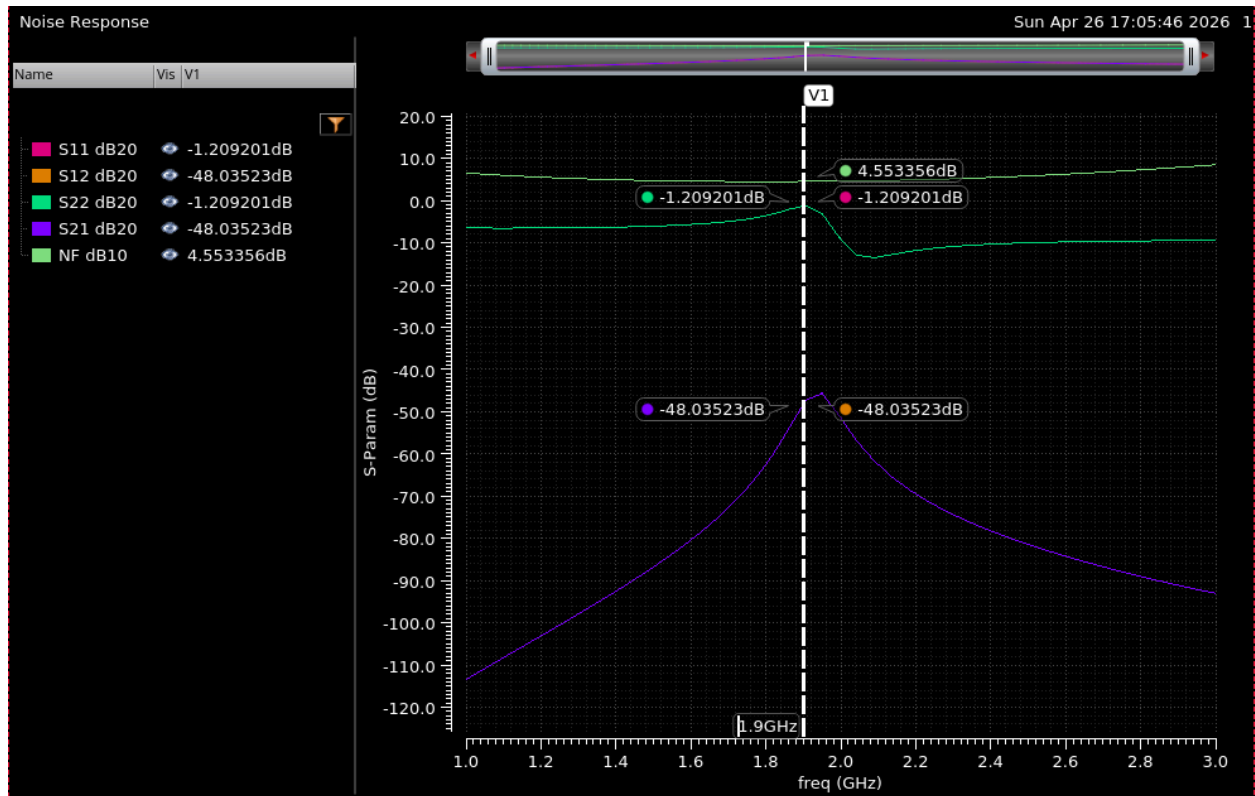
Doing dc analysis and finding the operating point:



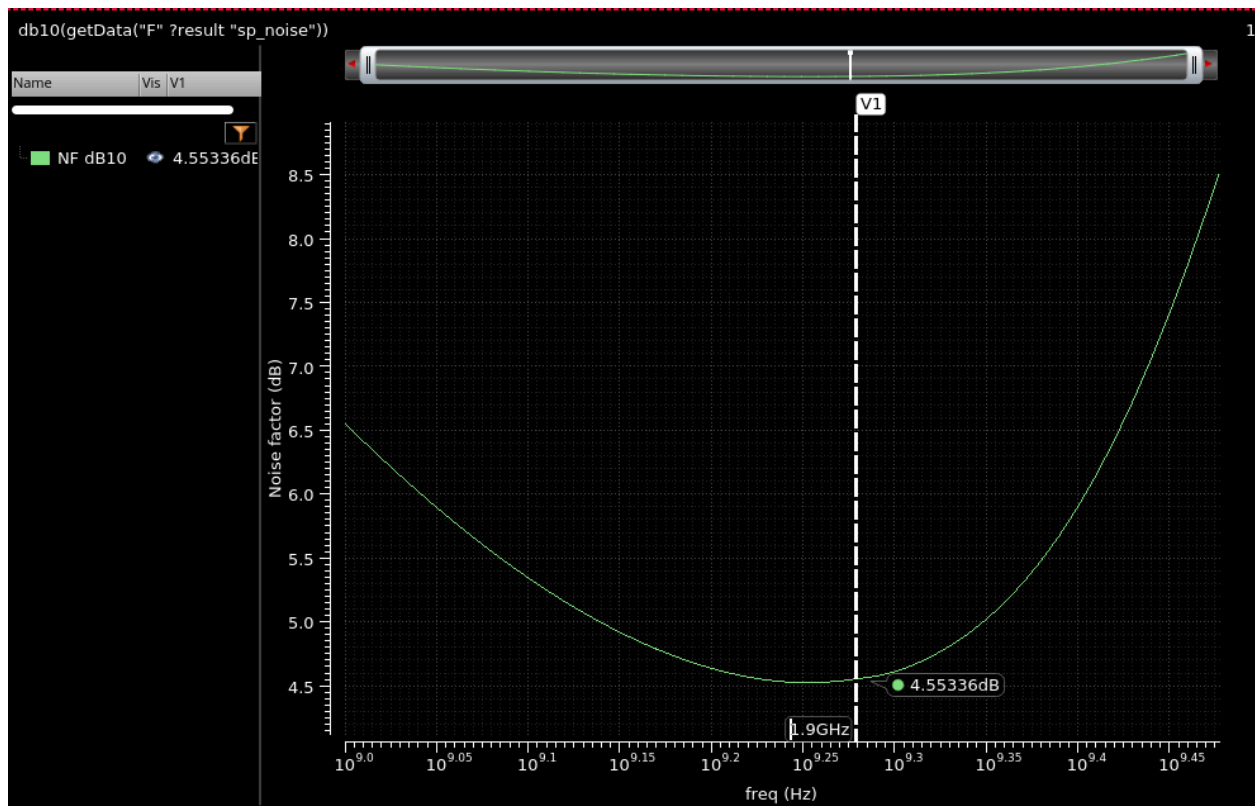
Operating point being=600mV

Tail Current being=1.995 mV

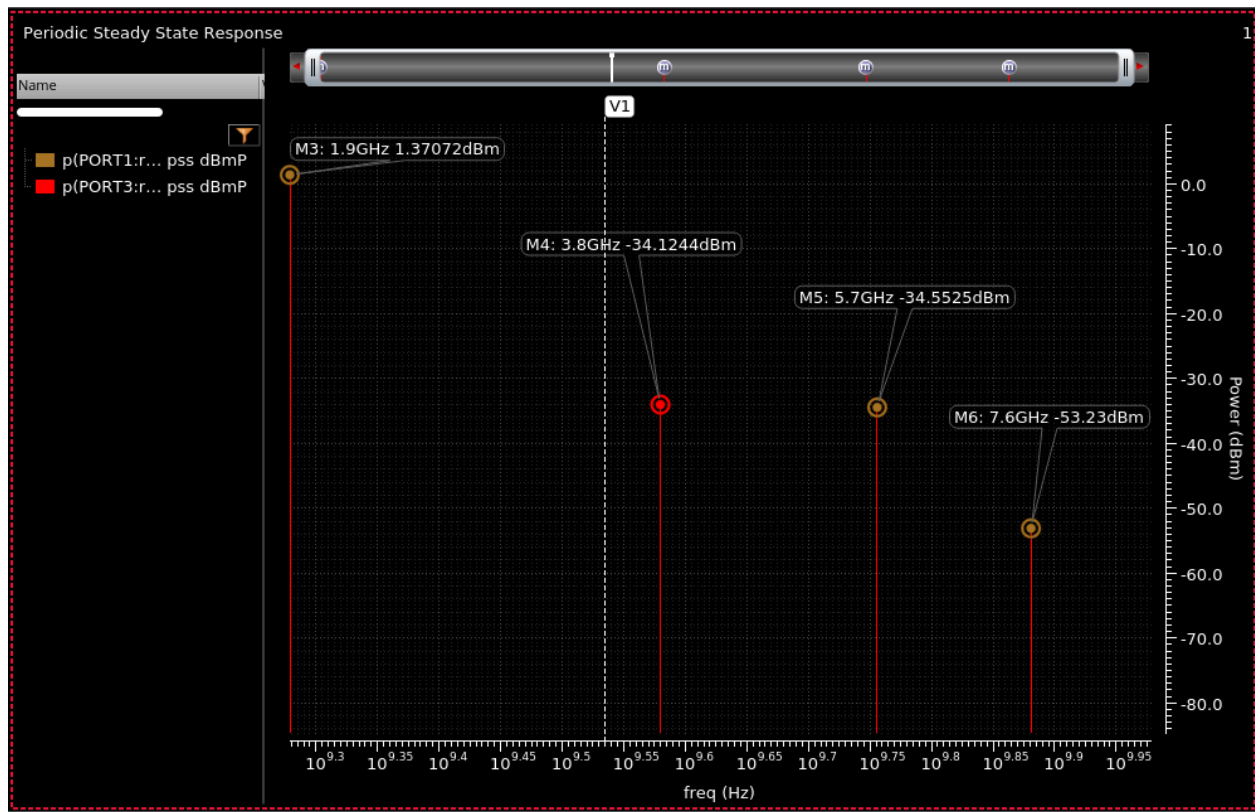
Graph showing S21(GAIN) AND the noise figure :



Noise figure:

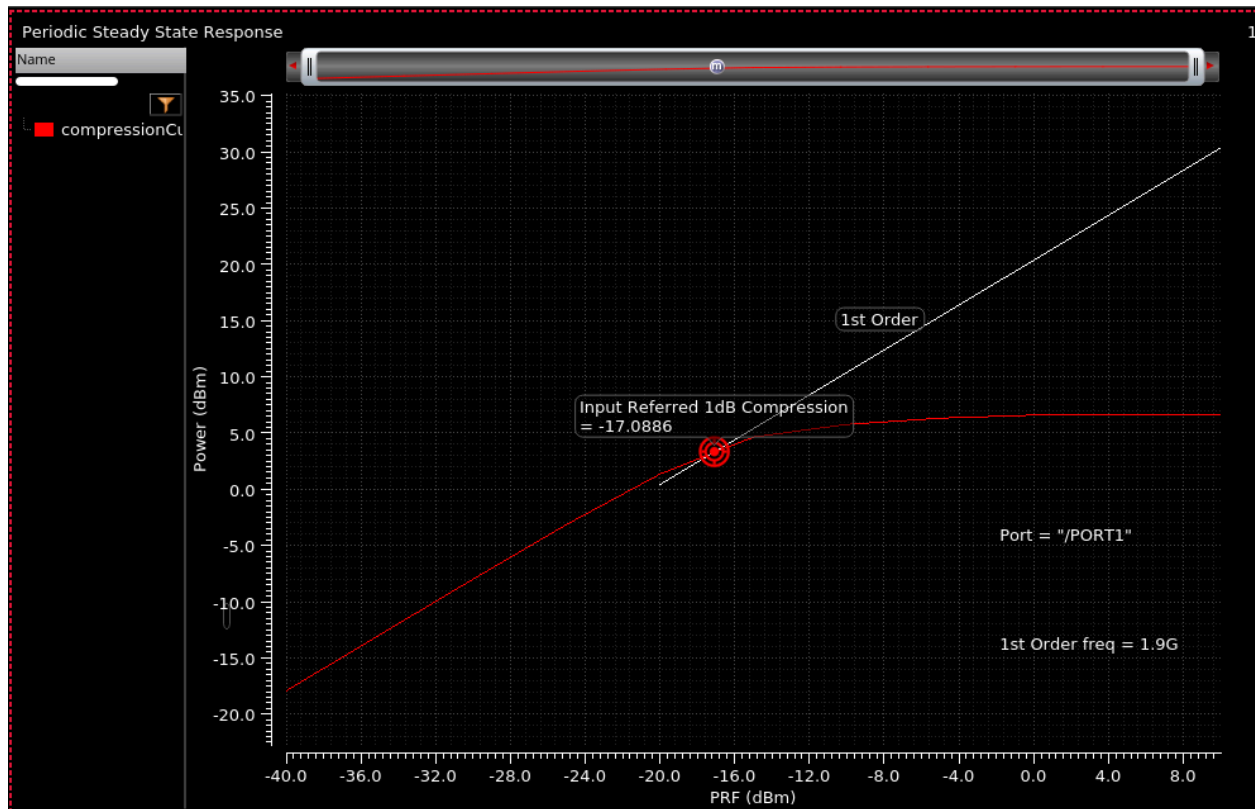


Pss simulation:



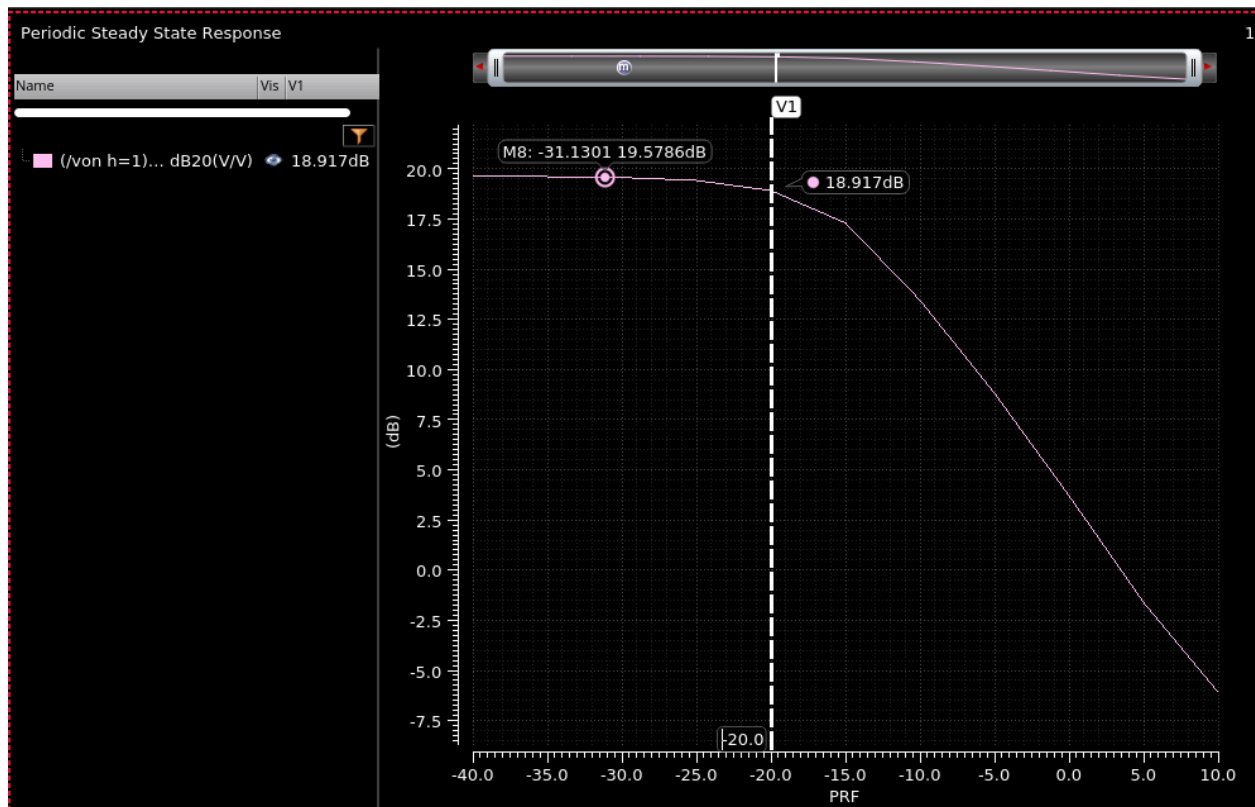
Spss simulation:

Differential setup: 1db compression point:

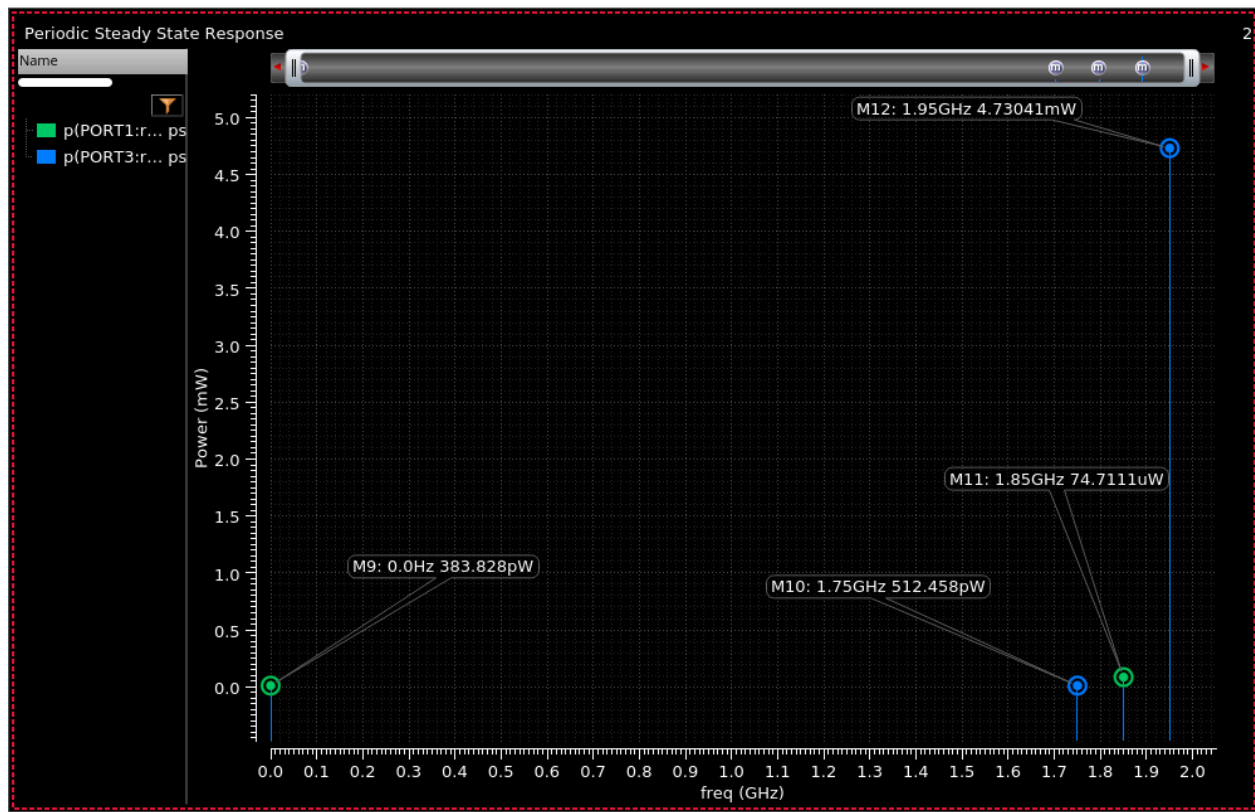


Compression point:-17.0886

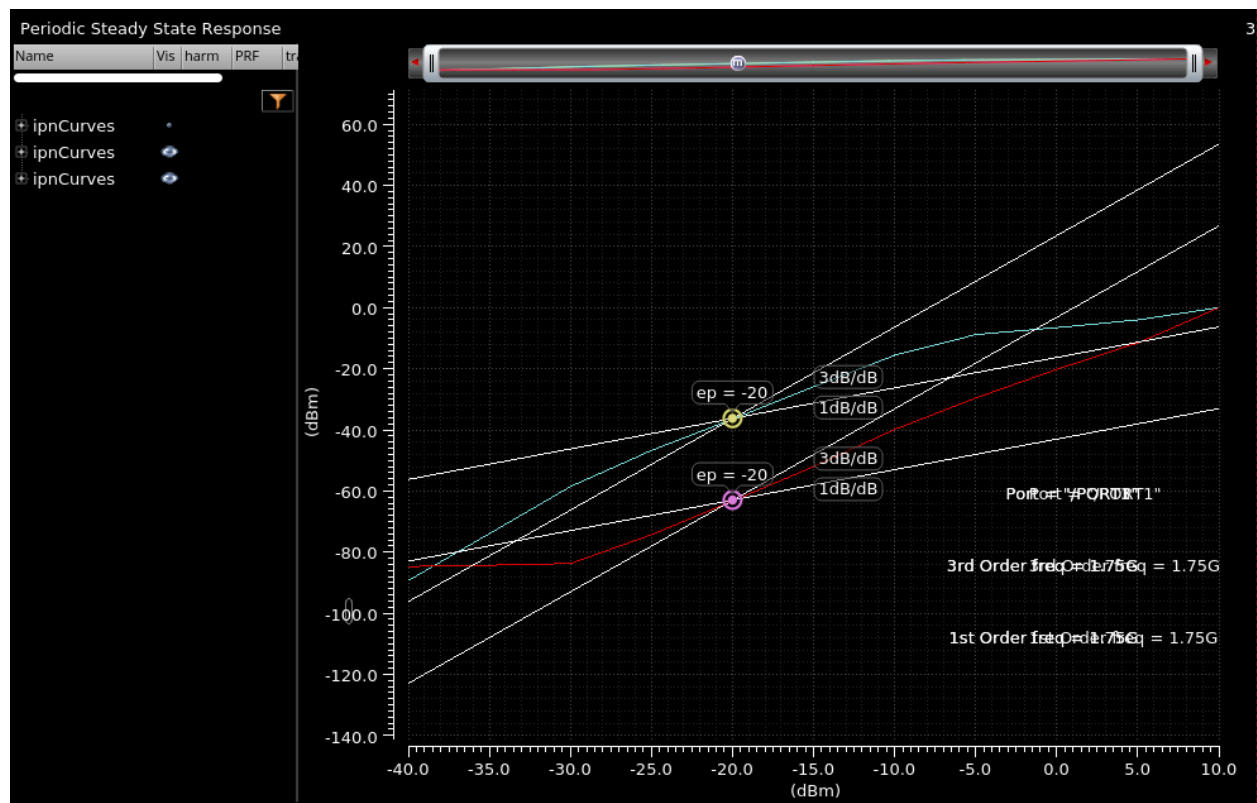
Voltage gain for single tone:



Two tone simulation:

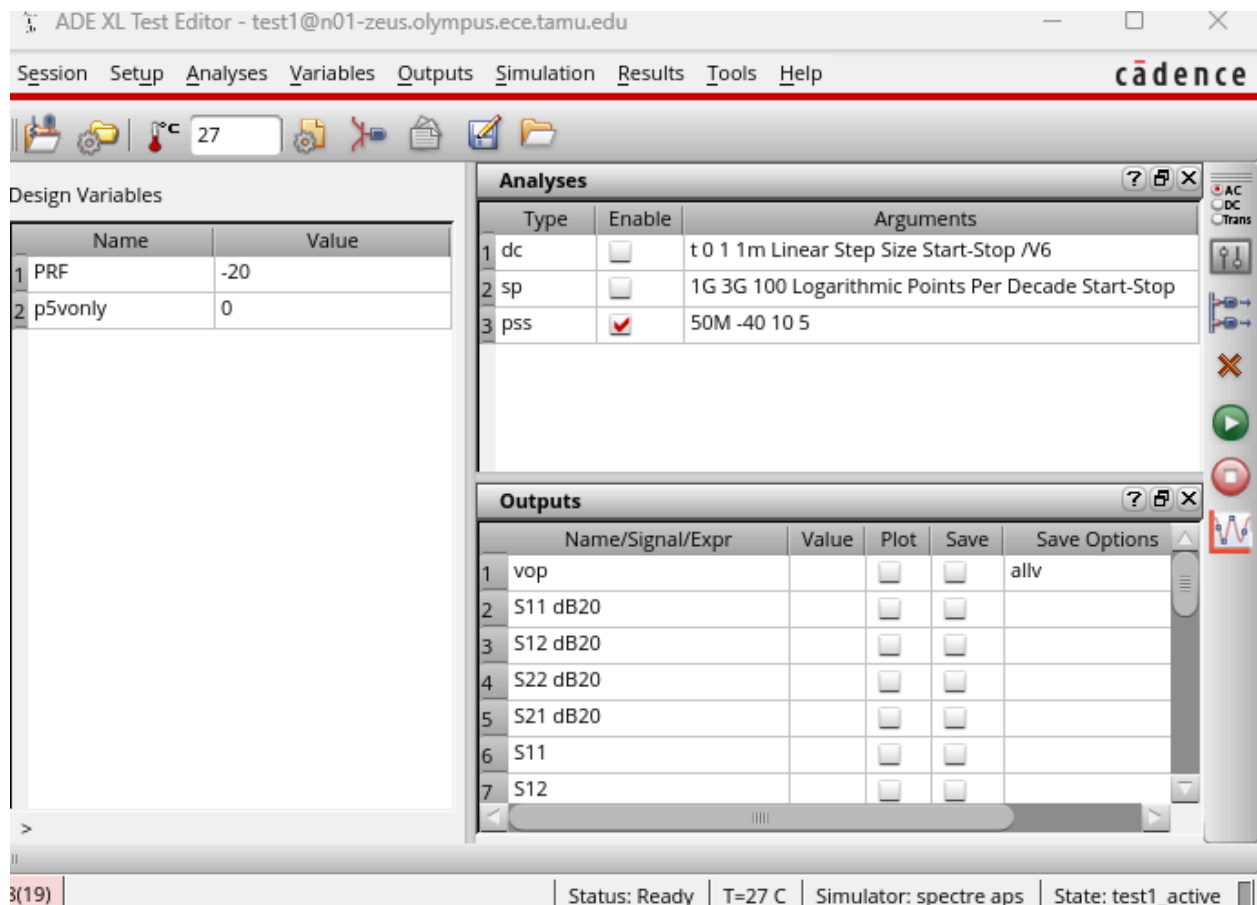


Two tone simulation spss:



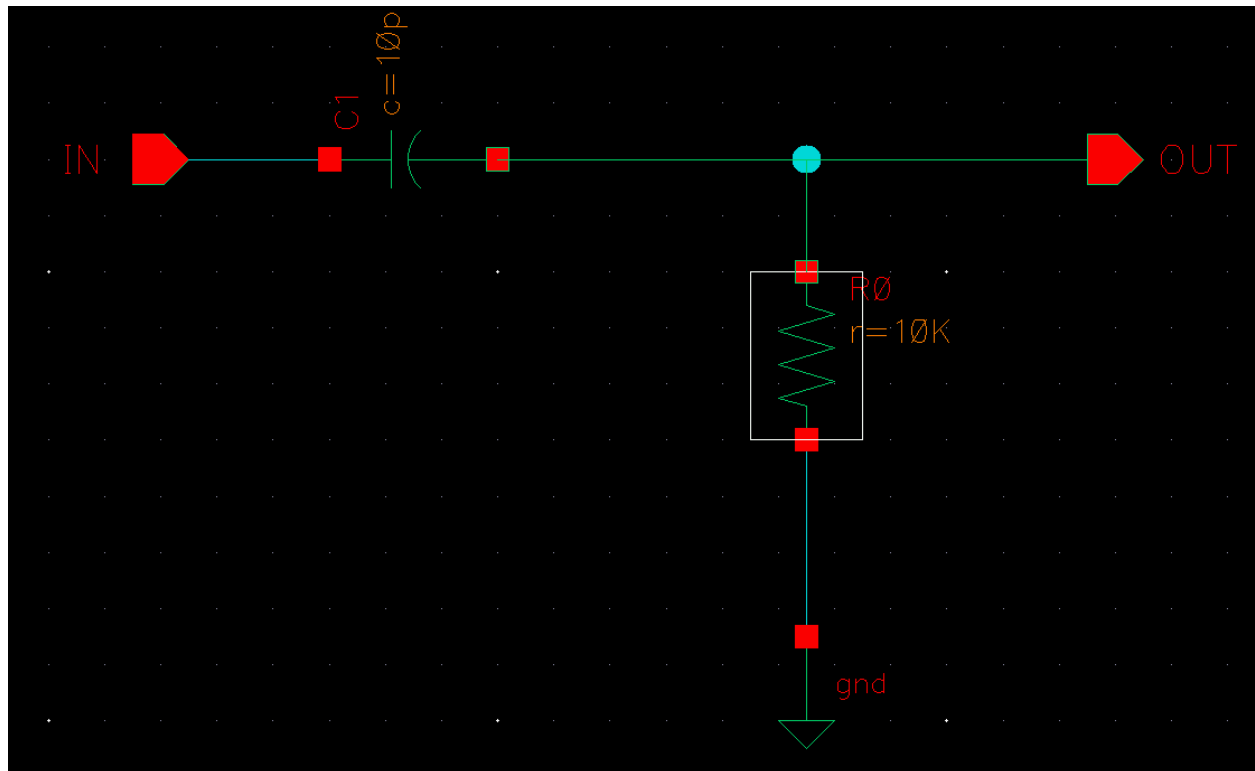
lip3 = -20

The ADEXL Setup Window:

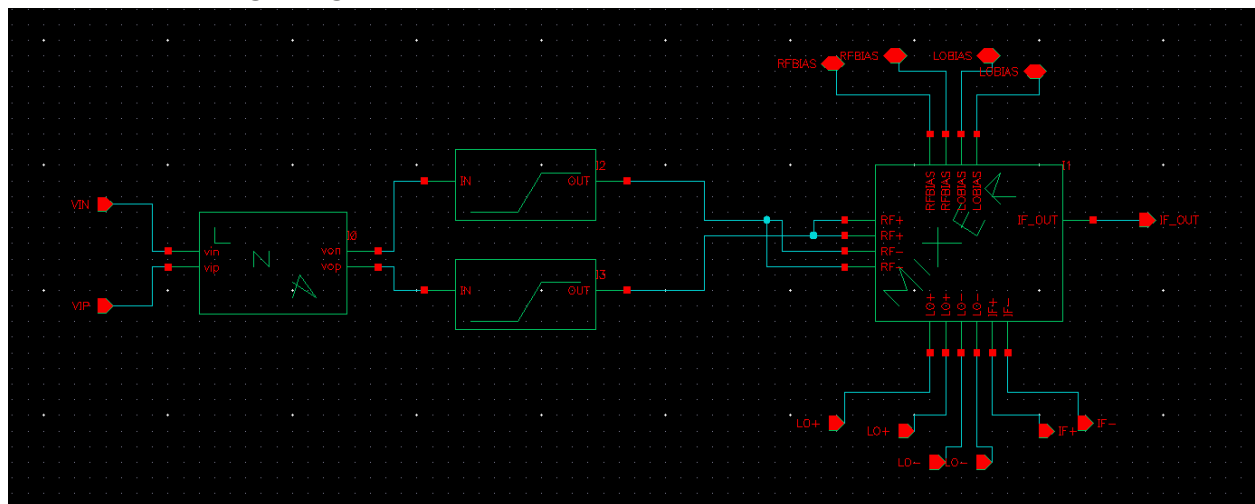


2. LNA and Mixer connection: Connect the differential output of the LNA to the RF port of the mixer designed in Lab 5 through an RC coupling network. Choose R in the order of Kohms and C in the order of pF. Notice that the RF transistors of the mixer will be biased through the resistors. How would they affect the NF? In this configuration, obtain the AC gain of the LNA. The peak of the AC response might have shifted from the simulation of the single-ended LNA due to the difference between the input parasitics of the buffer and the mixer. If this is the case, make appropriate adjustments to the load inductors of the LNA. Obtain the NF, Gain and IIP3 of your re-designed LNA. During this simulations, the LNA should remain connected to the mixer but the LO signal will be off

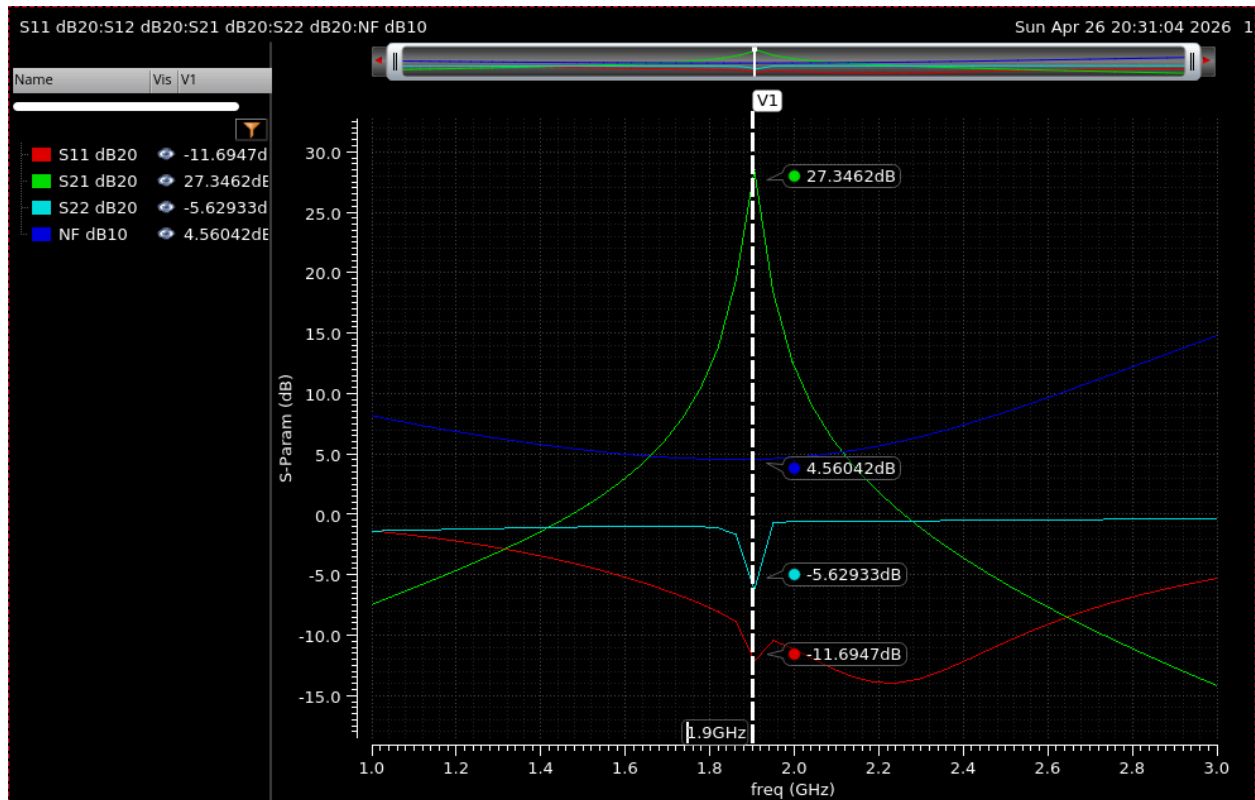
Low pass Filter:



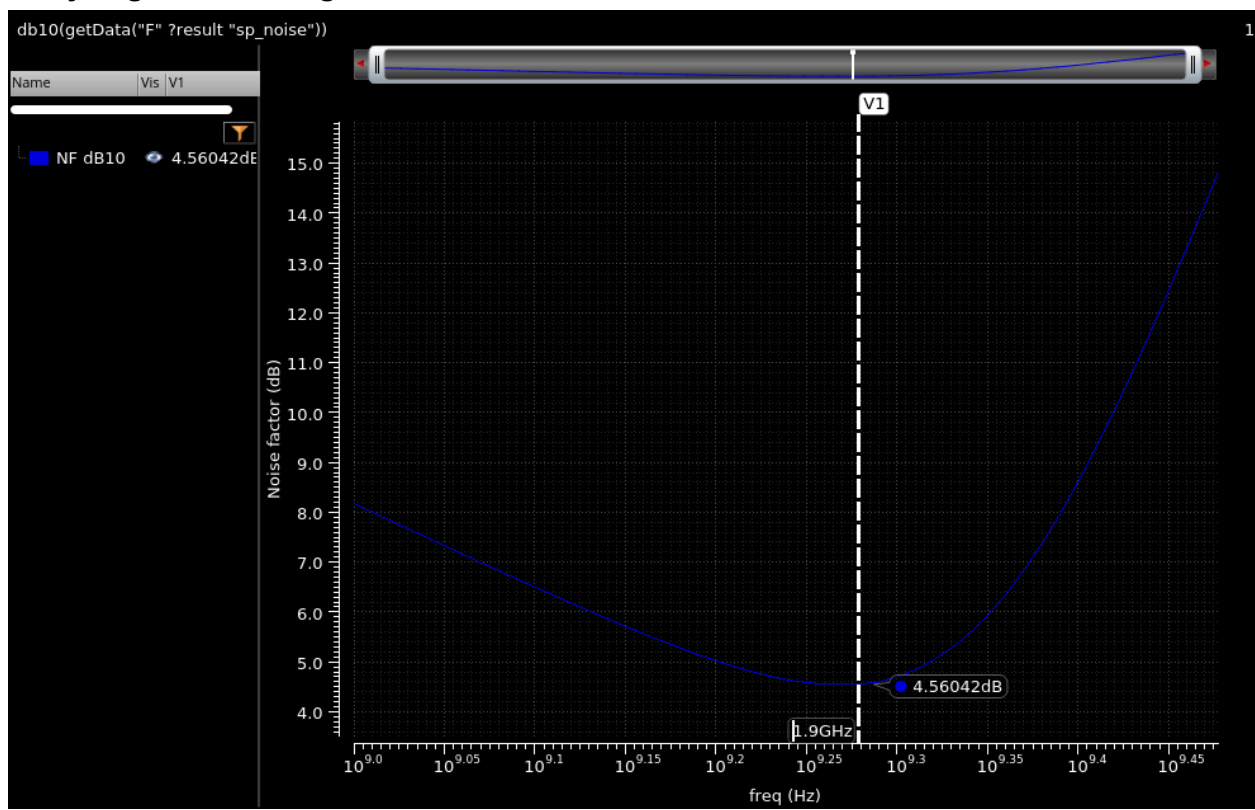
Schematic of integrating all of the blocks:



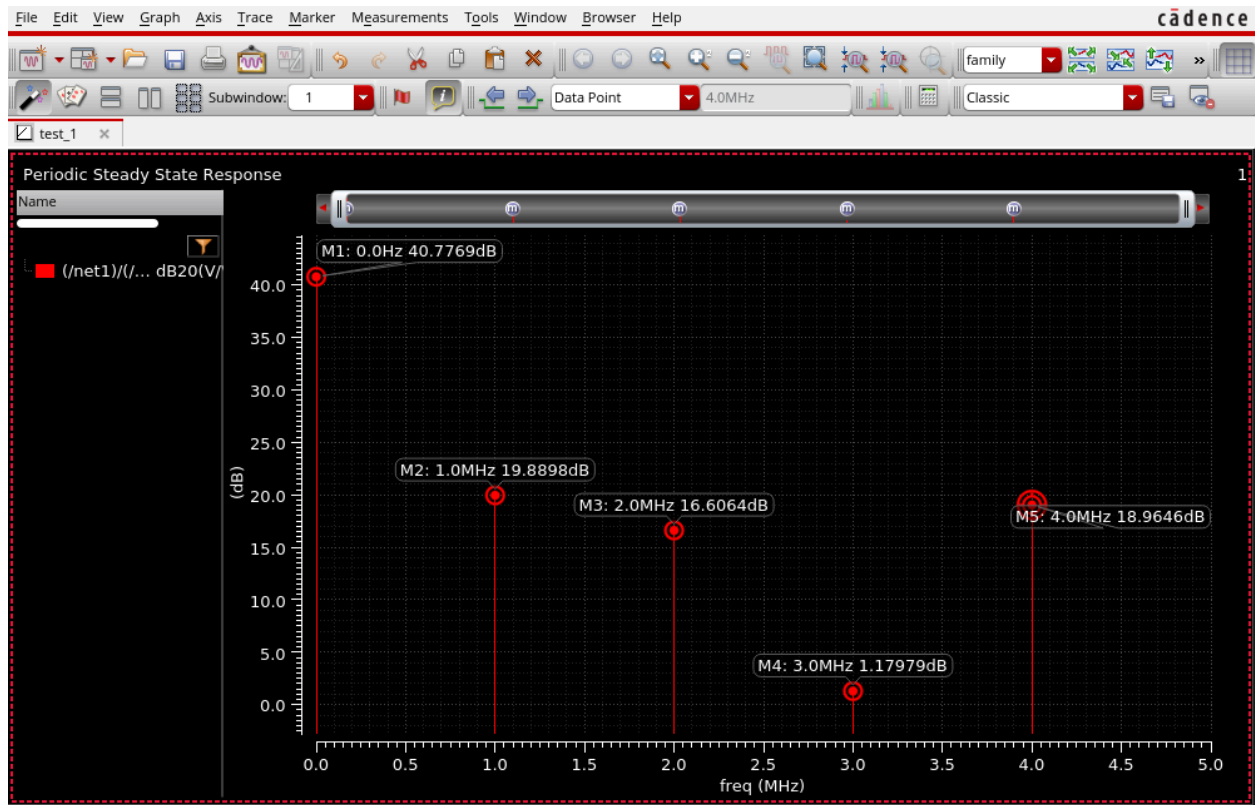
There is a shift in the s-parameter characters:
After connecting the blocks and analysing the LNA
SO The change is seen :



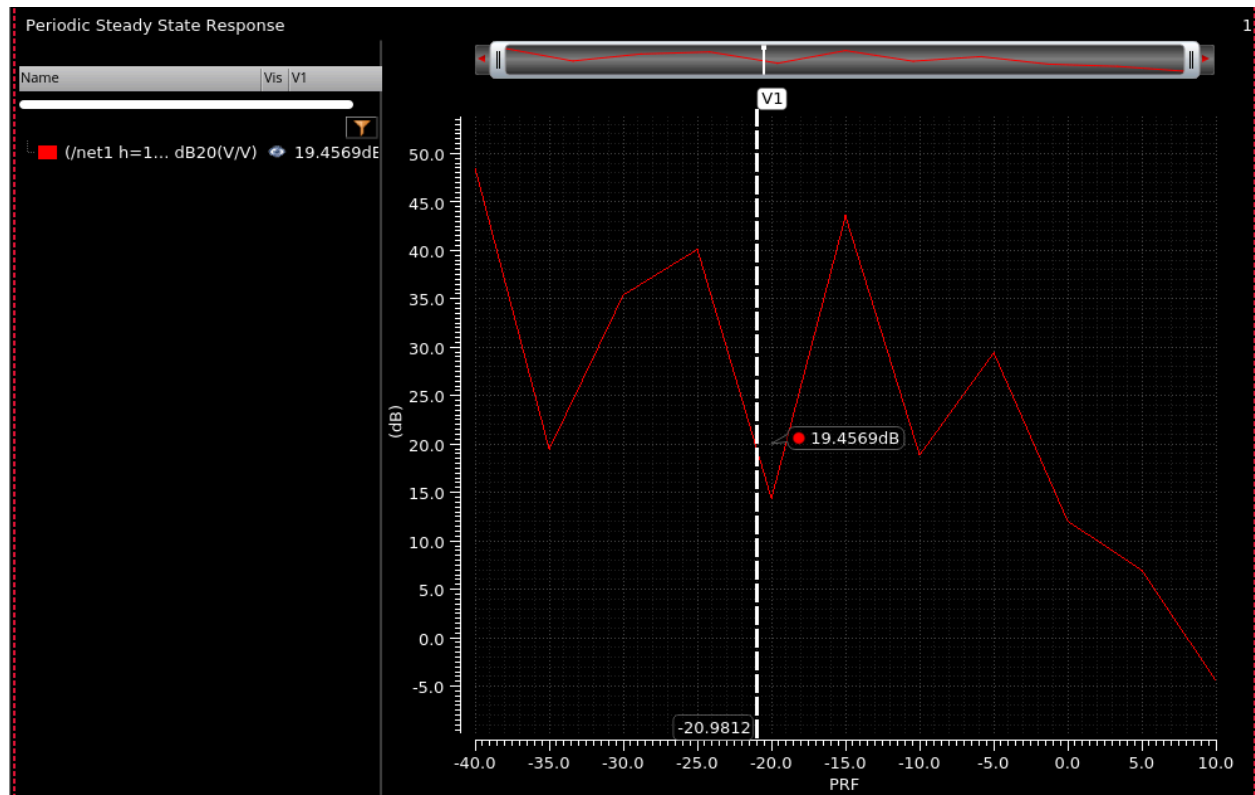
Analysing the Noise Figure:

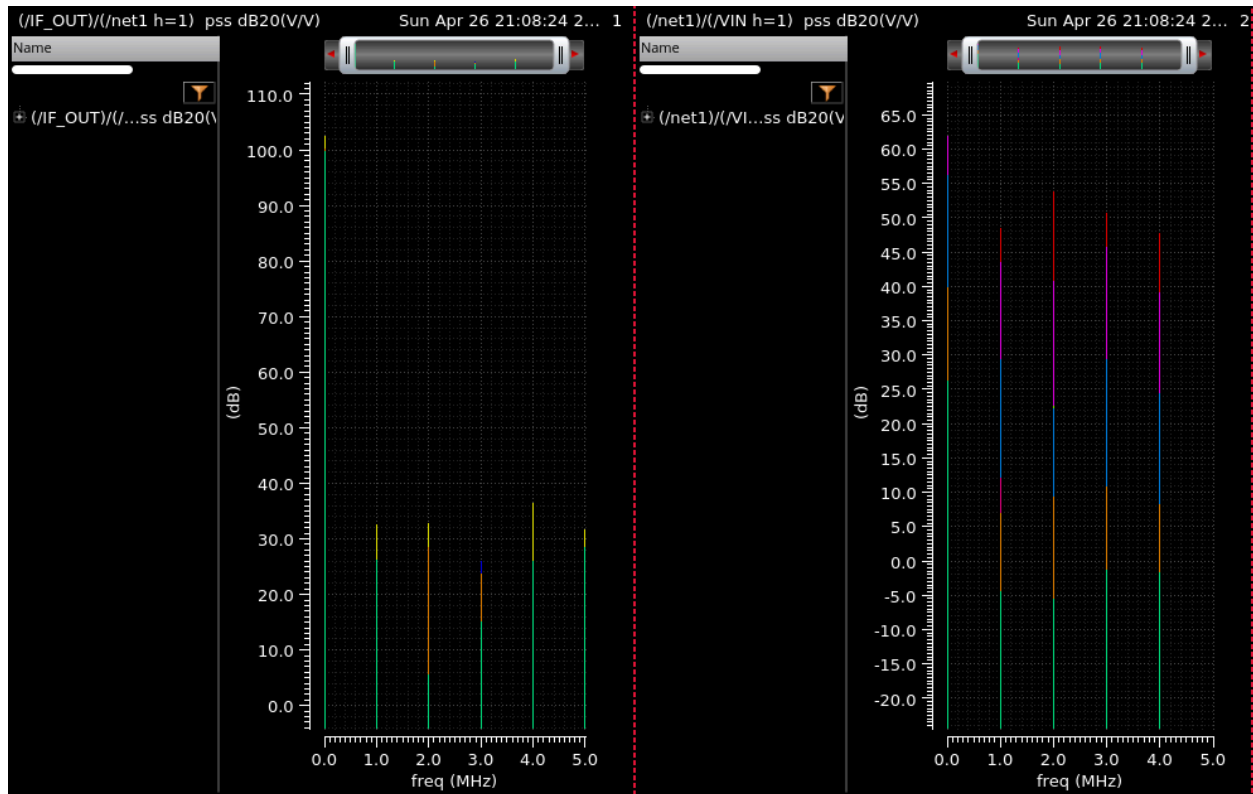


Gain:



SWEEPING AND GETTING THE GAIN:

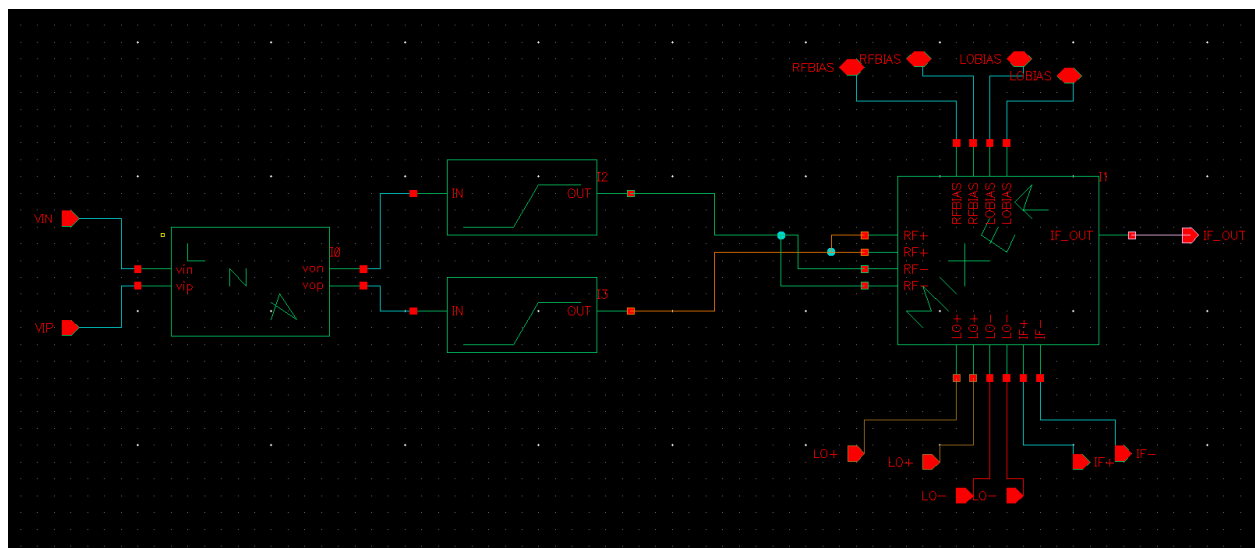
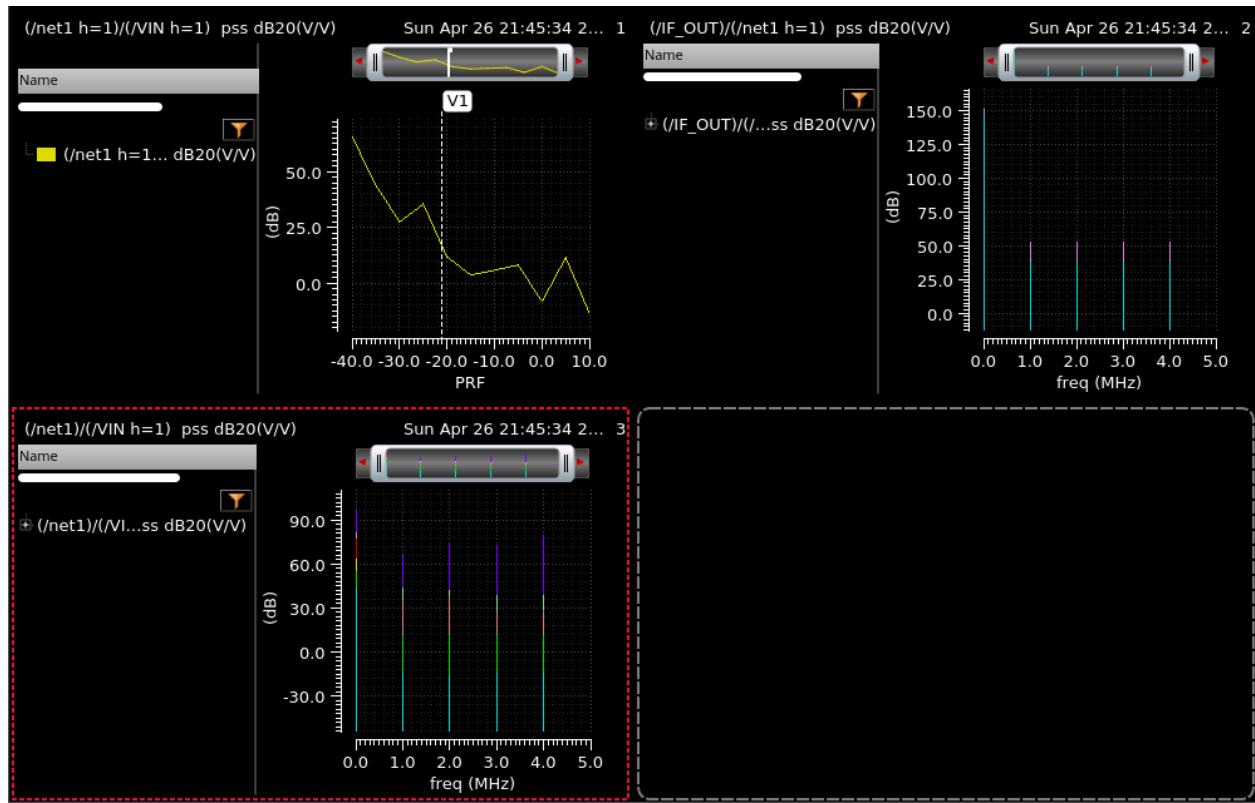




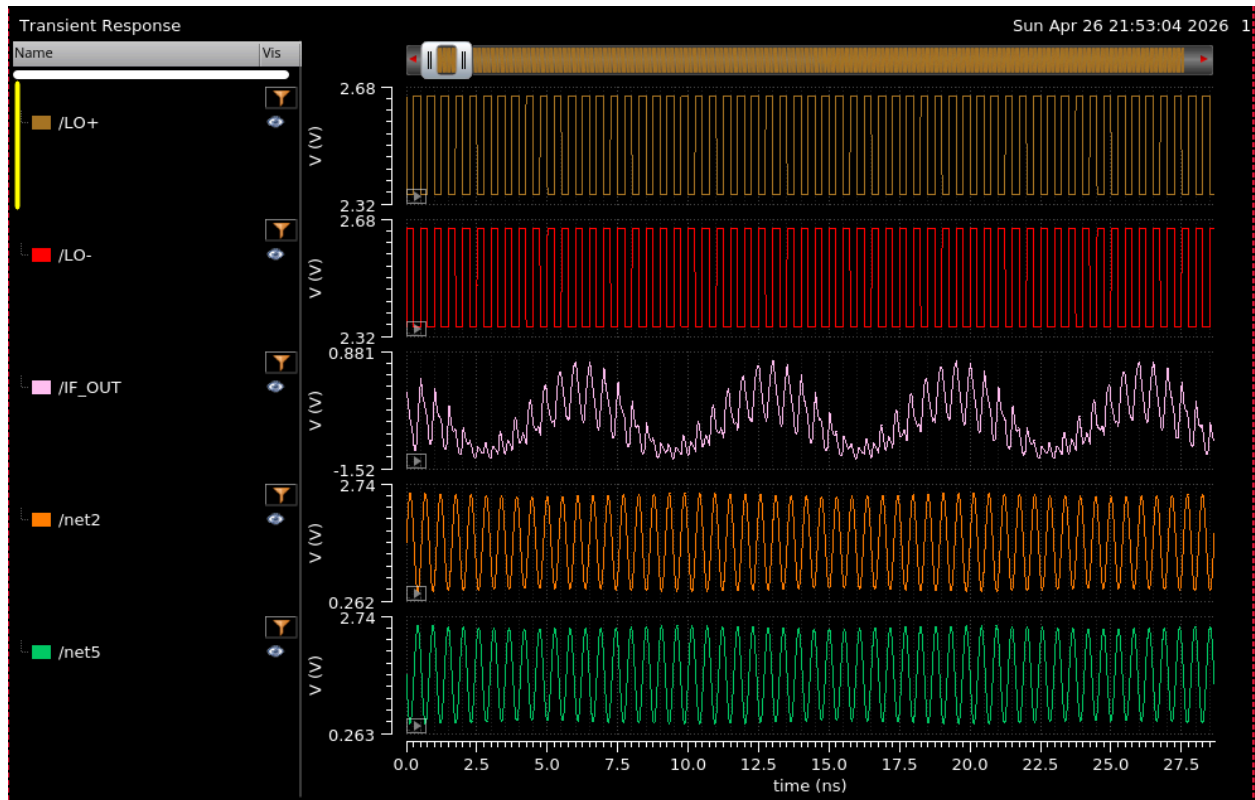
IIP3:

3. LNA and mixer simulation: Using the data of the performance of the mixer designed in Lab 5, calculate the expected cascaded (LNA+mixer) NF, gain and IIP3. Turn-on the LO signal (with the amplitude you used for the characterization of the mixer) and run a pss and/or transient simulation to verify the operation of the cascaded blocks. Measure the overall conversion gain.

Turning on the LO and then running the simulation:

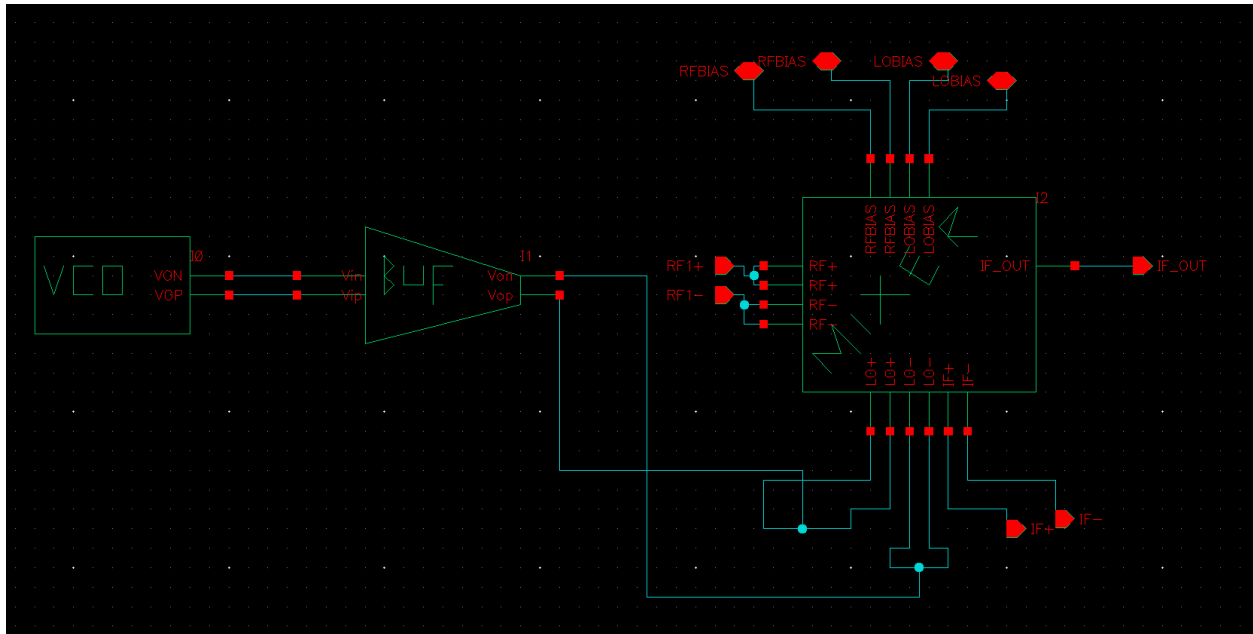


Transient analysis of all of the blocks:

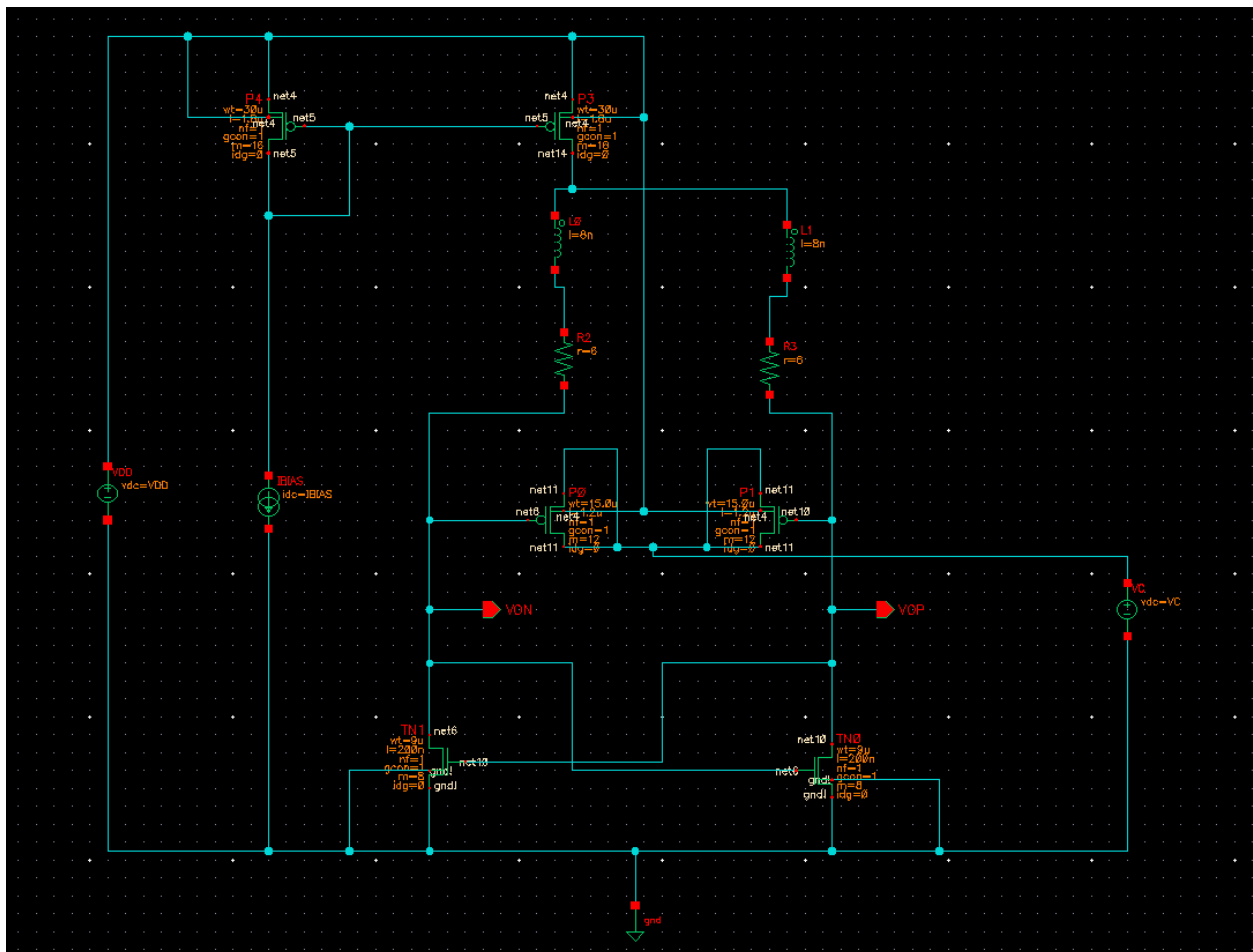


4.4. VCO and mixer connection and buffer design: On a separate schematic, place the VCO from Lab 6 and the mixer. The buffer to couple them will be a differential pair with an inductive/resistive load. Connect the VCO to the buffer (remove the 200fF capacitors that were employed to emulate parasitics), and the buffer to the LO port of the mixer. The RF port of the mixer should be biased but no signal should be applied. Design the buffer such that, for a 1.9GHz signal, the amplitude at the LO port is approximately the same that you used for the simulations in step

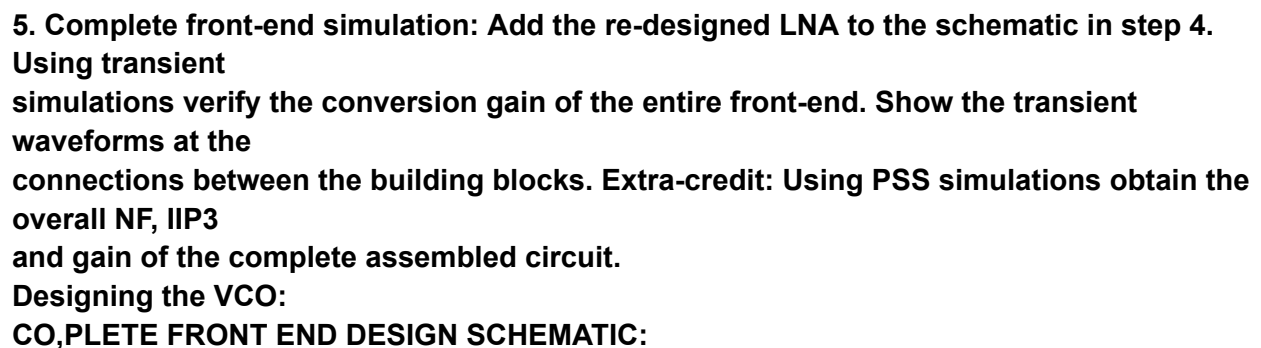
THE BLOCK DIAGRAM DEVELOPED:

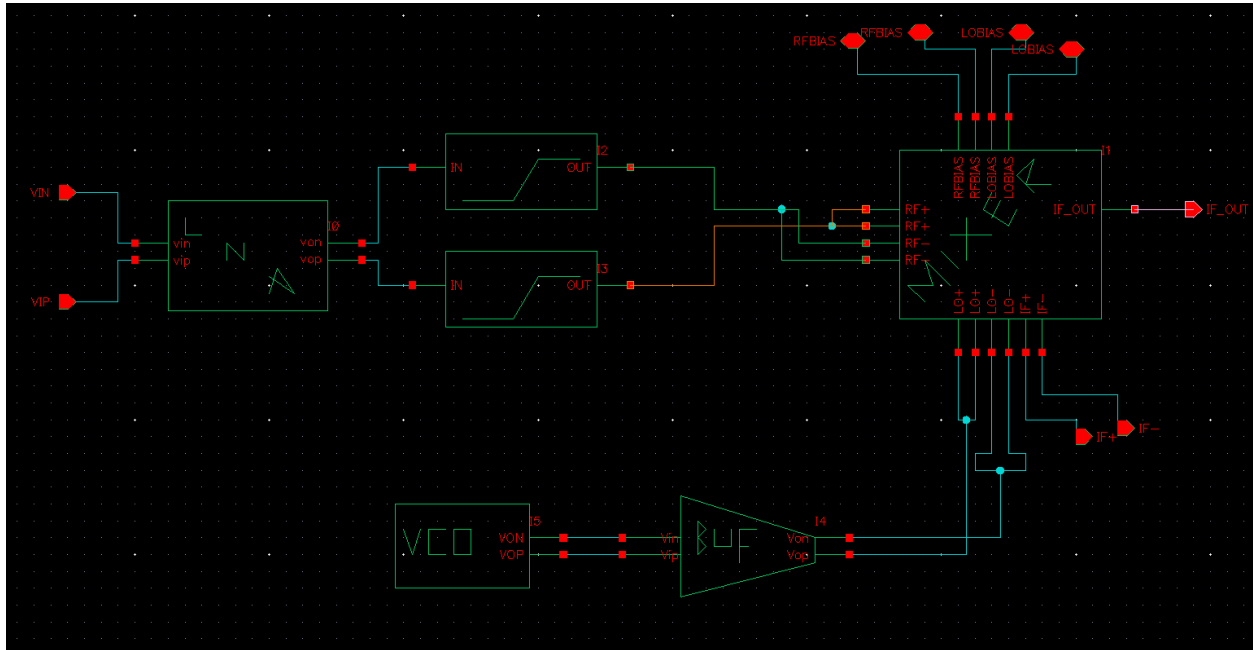


THE VCO:

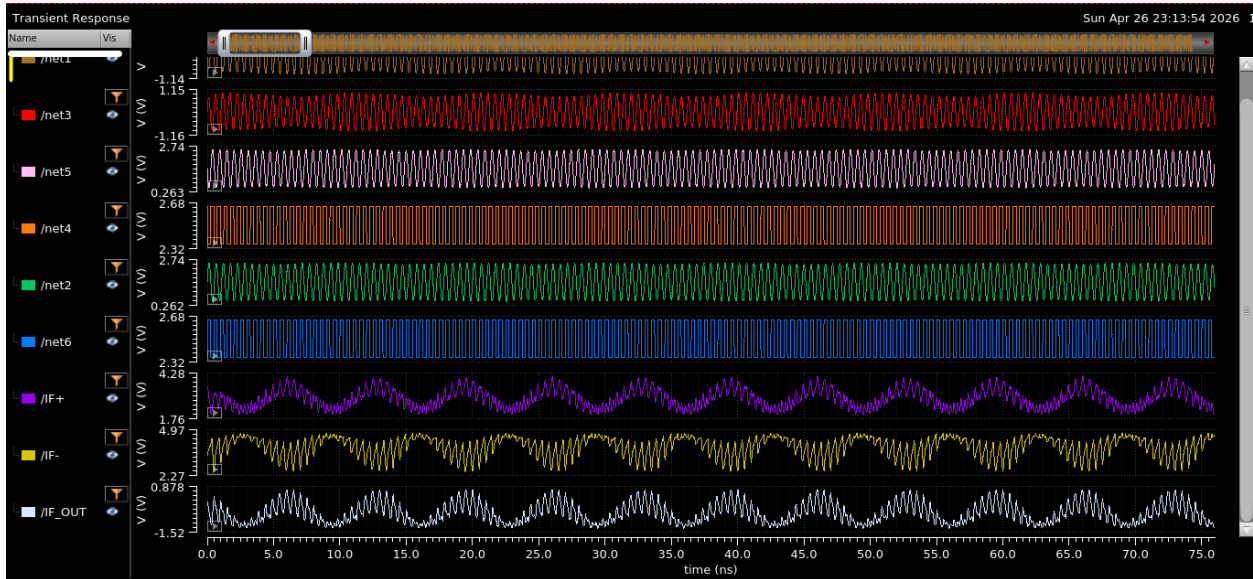


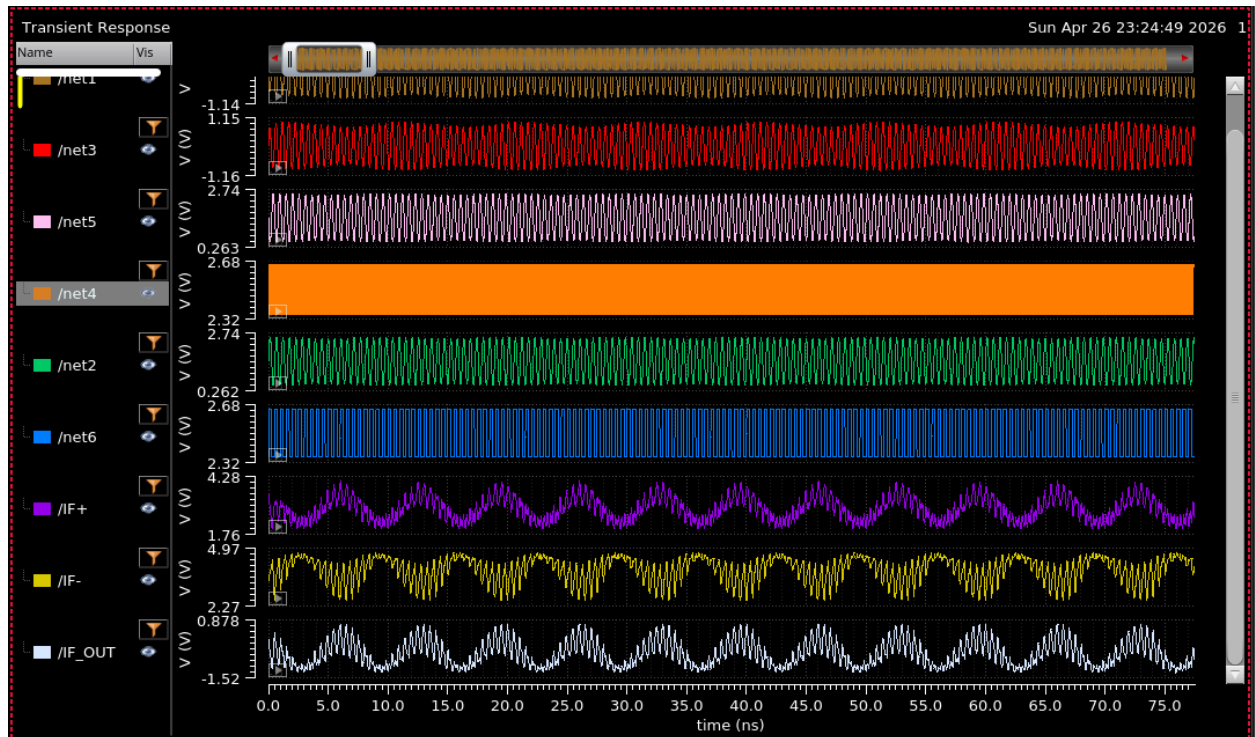
THE BUFFER:





FINAL TRANSIENT ANALYSIS:





Running the pss simulation:

Session Setup Analyses Variables Outputs Simulation Results Tools Help

cadence

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Design Variables

Name	Value
1 IBIAS	1m
2 VDD	3
3 VC	2
4 PRF	-20
5 p5vonly	0

Analyses

Type	Enable	Arguments
1 sp	☐	1G 3G 100 Logarithmic Points Per Decade Start-Stop
2 pss	☒	1M 5 -40 10 5
3 tran	☐	0 1u

Outputs

Name/Signal/Expr	Value	Plot	Save	Save Options
1 S11 dB20		☐	☐	
2 S12 dB20		☐	☐	
3 S21 dB20		☐	☐	
4 S22 dB20		☐	☐	
5 NF dB10		☐	☐	
6 (/IF_OUT)/(/net1 h=1) pss ...		☐	☐	
7 (/net1)/(/VIN h=1) pss dB2...		☐	☐	

> Results in ...r_shes123/test_1/simulation/integ

Additional pss simulation:

Part 2. MATLAB Integration

Procedure:

1. For your LNA and Mixer, you will first need to export your

S parameters vs. Frequency,

NF vs. Frequency,

Output Power vs. Input Power

Simulation results to a file in Cadence. You can use the Print command from the Results Menu in Analog

Environment to view the desired outputs. Figure 1. shows the 2 port S parameter results window. You should

change the data format to Engineering (Exponent) from the Display Options Menu in the Results Display

Window under Expressions->Display Options.

Fig.1. 2 Port S Parameter Results Display in Cadence

Export this file to a .txt file from the Window Print Menu.

2. In MATLAB Simulink RF Blockset, you will use General Amplifier instead of

S-Parameters Amplifier that

you used in Lab2. Read the “Amp File Format” section from MATLAB Help to understand the .amp file

format.

You can read the results files generated by Cadence, in MATLAB by using the “dlmread” command.

Note that you will need to convert the polar format that Cadence uses into rectangular format that MATLAB

uses for complex parameters.

After reading the circuit parameters from Cadence generated file, you will need to create a valid .amp format.

You can use the .amp file you generated in the “General Amplifier” block and use it in your Simulink simulation.

3. By using the Visualization tab of the Physical blocks, plot the S parameters, NF and Nonlinearity of your

LNA and Mixer in MATLAB Simulink. Figure 2. shows the S parameter plots for the LNA design given in

Lab 3, obtained from General Amplifier block in MATLAB.

Fig.2 LNA S parameters in Cadence, exported from Cadence into a General Amplifier Block

4. Combine your LNA and Mixer, you can use an ideal LO to represent the oscillator and disregard the phase

noise of the oscillator. Perform the complete RF Front End Simulation, provide plots of this simulation. How

do the results compare with Cadence results, are they similar? Discuss

LNA Simulation Results — Cadence to MATLAB Export

Cadence S-Parameter Simulation at 1.9 GHz

Figure 1 shows the simulated S-parameters and Noise Figure (NF) of the LNA obtained from Cadence. The cursor is placed at the target operating frequency of 1.9 GHz, and the key performance metrics are annotated on the plot.

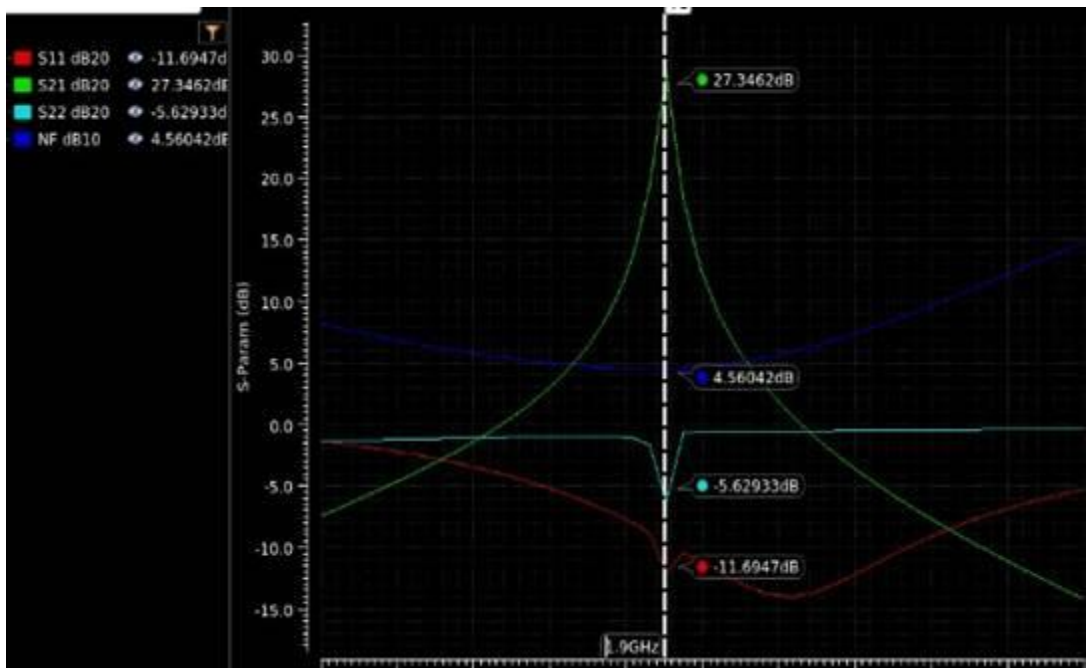


Figure 1. LNA S-Parameter and NF Simulation Results in Cadence at 1.9 GHz

Summary of Key Parameters at 1.9 GHz

Table 1 summarizes the LNA performance metrics extracted from the Cadence simulation at the 1.9 GHz design frequency.

Parameter	Symbol	Value	Unit
Forward Gain	S21	27.35	dB
Input Return Loss	S11	-11.69	dB
Output Return Loss	S22	-5.63	dB
Noise Figure	NF	4.56	dB

Table 1. LNA Key Performance Metrics Extracted from Cadence at 1.9 GHz

Analysis of Simulation Results

Forward Gain ($S_{21} = 27.35$ dB)

The LNA achieves a forward gain of 27.35 dB at 1.9 GHz, confirming that the design is well-tuned to the target frequency. The gain curve rises steeply and peaks at 1.9 GHz, which is consistent with the expected narrowband LNA behavior. This level of gain is sufficient to suppress the noise contribution of subsequent stages in the receiver chain.

Input Return Loss ($S_{11} = -11.69$ dB)

The input return loss of -11.69 dB is below the commonly accepted threshold of -10 dB, indicating good input impedance matching at 1.9 GHz. This means that most of the input signal power is absorbed rather than reflected, which is critical for minimizing signal loss and maintaining system sensitivity.

Output Return Loss ($S_{22} = -5.63$ dB)

The output return loss of -5.63 dB does not meet the preferred -10 dB threshold, suggesting that the output port matching is not optimal. This could lead to some signal reflection at the LNA output, which may affect the performance of cascaded stages such as the mixer. Further optimization of the output matching network is recommended in future design iterations.

Noise Figure (NF = 4.56 dB)

The noise figure of 4.56 dB at 1.9 GHz is within an acceptable range for this design stage. While practical low-noise amplifier designs typically target NF values below 2–3 dB, the result reflects inherent trade-offs between gain and noise performance. In a full receiver chain analysis, the Friis formula confirms that the LNA NF dominates overall system noise figure, emphasizing the importance of minimizing it in future design refinements.

Exporting Cadence Results to MATLAB (.amp Format)

The simulation results were exported from Cadence as a .txt file and converted into a valid .amp format for use in the MATLAB Simulink RF Blockset General Amplifier block. The following procedure was followed:

- The S-parameters were exported from Cadence in polar format (magnitude and phase angle).
- The polar data was converted to rectangular (complex) format in MATLAB using: $S_{\text{rect}} = |S| \times e^{j\theta}$, where θ is expressed in radians.
- The dlmread command was used in MATLAB to import the Cadence-generated .txt file.
- A valid .amp file was then constructed with the appropriate header and data format required by the General Amplifier block in Simulink.

Comparison of Cadence and MATLAB Simulink Results

After importing the .amp file into the General Amplifier block in MATLAB Simulink, the S-parameters, NF, and nonlinearity were visualized using the Visualization tab of the Physical block. The MATLAB results closely matched the Cadence simulation, as summarized below:

- The gain (S21), input/output return loss (S11, S22), and NF curves showed strong agreement between both environments at 1.9 GHz.
- Minor discrepancies in gain flatness and NF near band edges were observed, attributable to MATLAB's interpolation across the discrete frequency points exported from Cadence.
- The overall agreement confirms that the .amp file export and import procedure was performed correctly, and that the General Amplifier block is an accurate representation of the Cadence-designed LNA.

The MATLAB RF Front End simulation, combining the LNA and Mixer (with an ideal LO), produced output results consistent with the expected cascade performance derived from the individual Cadence component results